Project Summary Report

Project 3

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Executive Summary

With extensive growth in the central Texas region, the Highland Lakes serve as a vital source of water for much of the region. The Highland Lakes are a chain of dammed water reservoirs along the Colorado River. The Lower Colorado River Authority (LCRA) manages the dams and the hydropower off them. The LCRA officials want a change detection analysis done for the Highland Lakes throughout the years. For this study, the two years picked for the before and after are 1987 and 2011.

To document this change, the land cover needs to be first classified. Five classifications with the designated codes are used in this study are: (0) Water, (1) Woodland, (2) Grassland, (3) Crops, (4) Developed, and (5) Barren. Training samples were then created. For this study, 30 training samples were created for all five classifications.

Supervised classification was then performed where the user groups pixels with similar spectral signatures. The software then performed supervised classification based on the training samples on the Landsat images for both 1987 and 2011. The overall accuracy of the supervised classification for 1987 is 33.3% and for 2011, it's 43.3%.

Unsupervised classification was then performed where the software groups pixels with similar spectral signatures into classes defined by the user. A land cover classification is then assigned after analyzing each classification group for both 1987 and 2011. The overall accuracy of the supervised classification for 1987 is 53.3% and for 2011, it's 40.0%.

Visually, a vast difference can be seen between 1987 and 2011. The developed classification is much more prevalent in both the supervised and unsupervised model in 2011 as compared to 1987. More will be written about this difference in part 2.

Summary of Approach

Change detection analysis was done with the following tasks: collect and assimilate project data, analyze spectral signatures, classify land cover, and perform accuracy assessment

Collect

Data was gathered from the USGS's Earth Explorer. Landsat images for both 1987 and 2011 were extracted from this website after selecting the study area around Highland Lakes and project to NAD 1983 UTM Zone 14N to the project geodatabase.

Analyze Spectral Signatures

Five classifications with the designated codes are used in this study are: (0) Water, (1) Woodland, (2) Grassland, (3) Crops, (4) Developed, and (5) Barren. For this study, 30 random sample points were created. Spectral signatures were produced from these samples as shown in Figure 1.

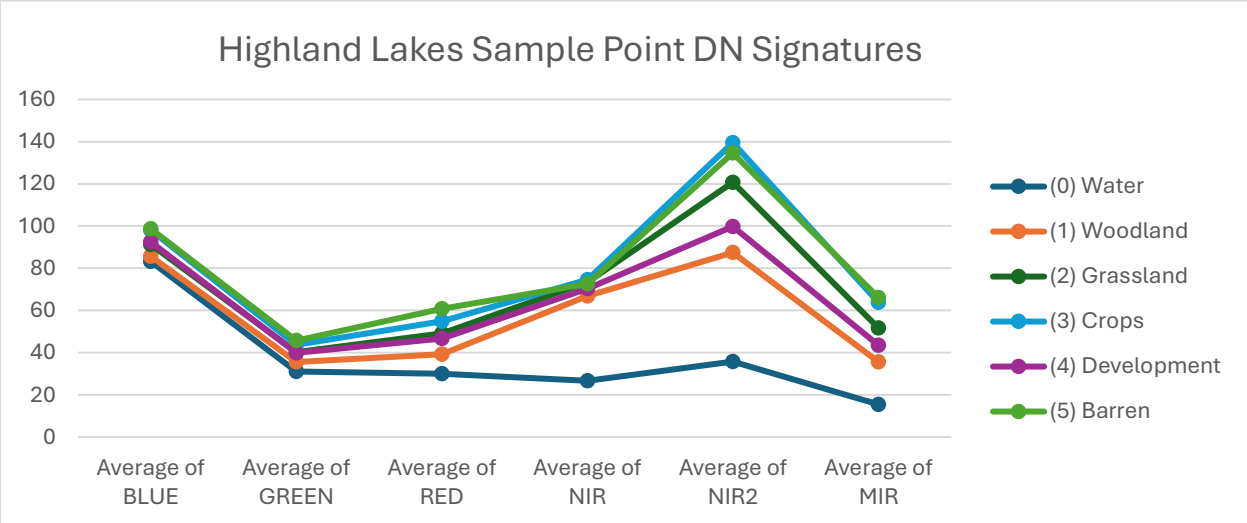


Figure 1

Classify Land Cover

Create a new schema based on the National Land Cover Database (2011) (NLCD2011) classification schema. The new schema will be based on the five classification codes stated above. This classification schema will be saved and used for the rest of the project. Training samples were then created for all five of the land cover classes for this project.

Supervised classification was then performed where the user groups pixels with similar spectral signatures. The software then performed supervised classification based on the training samples on the Landsat images for both 1987 and 2011. Figure 2 and Figure 3 show the years 1987 and 2011 respectively.

Unsupervised classification was then performed where the software groups pixels with similar spectral signatures into classes defined by the user. A land cover classification is then assigned after analyzing each classification group for both 1987 and 2011. Figure 4 and Figure 5 show the years 1987 and 2011 respectively.

Accuracy Assessment

This is covered in the Executive Summary and Accuracy Assessment section of this document.

Results

The following maps show the results for the supervised and unsupervised model analysis. The increase in developed land is visibly clear here between Figure 2 and Figure 3 plus for Figure 4 and Figure 5.

Highland Lakes 1987

Supervised Classification

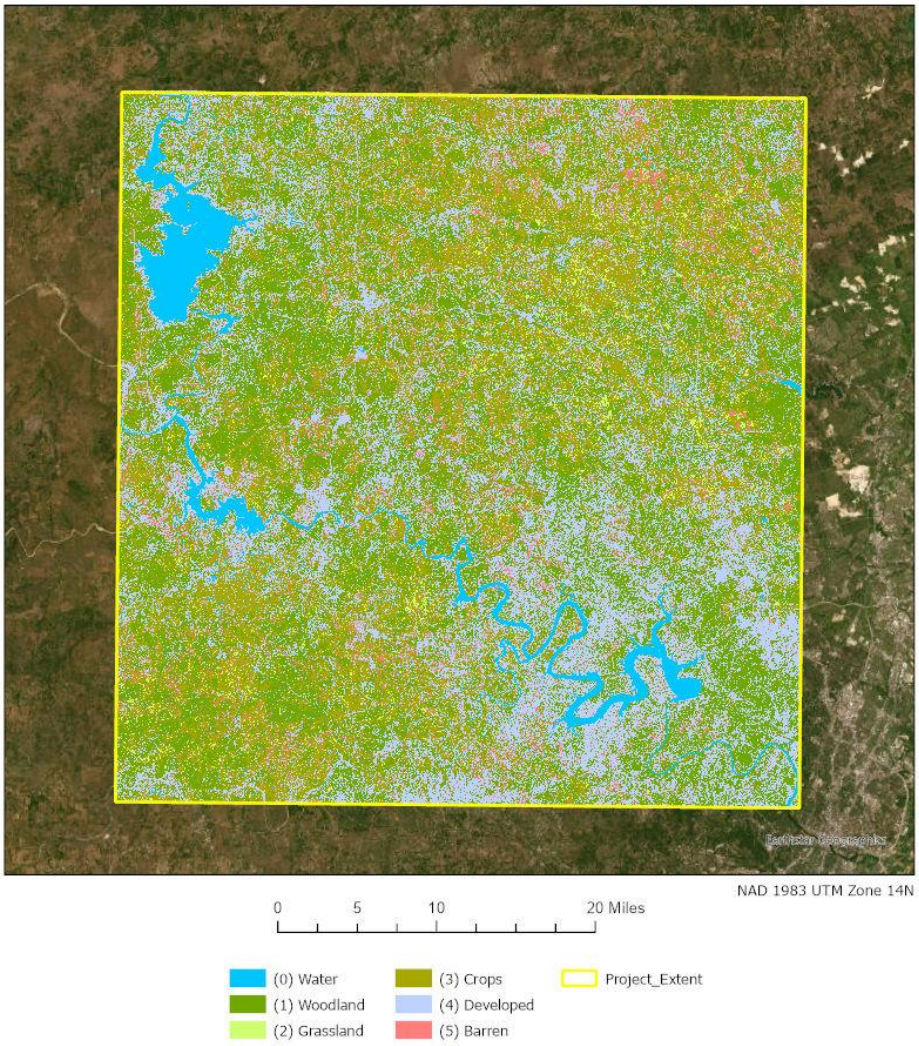


Figure 2

Highland Lakes 2011
Supervised Classification

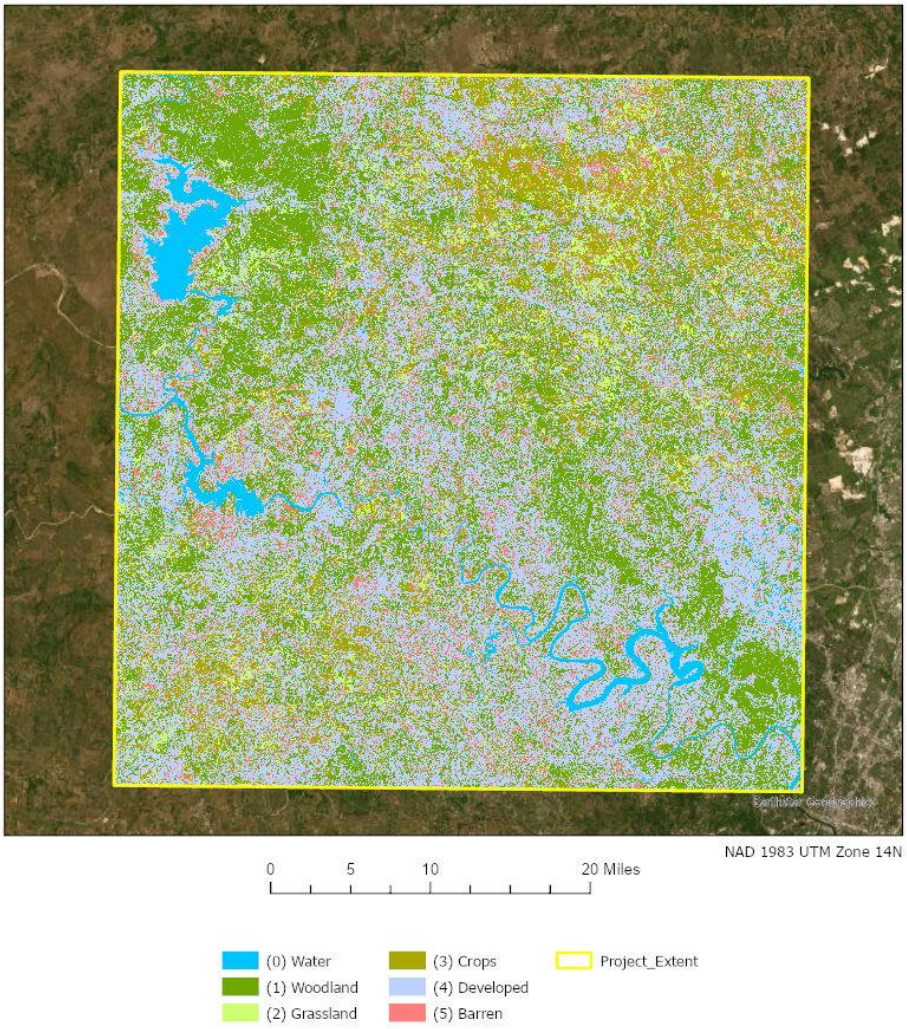


Figure 3

Highland Lakes 1987

Unsupervised Classification

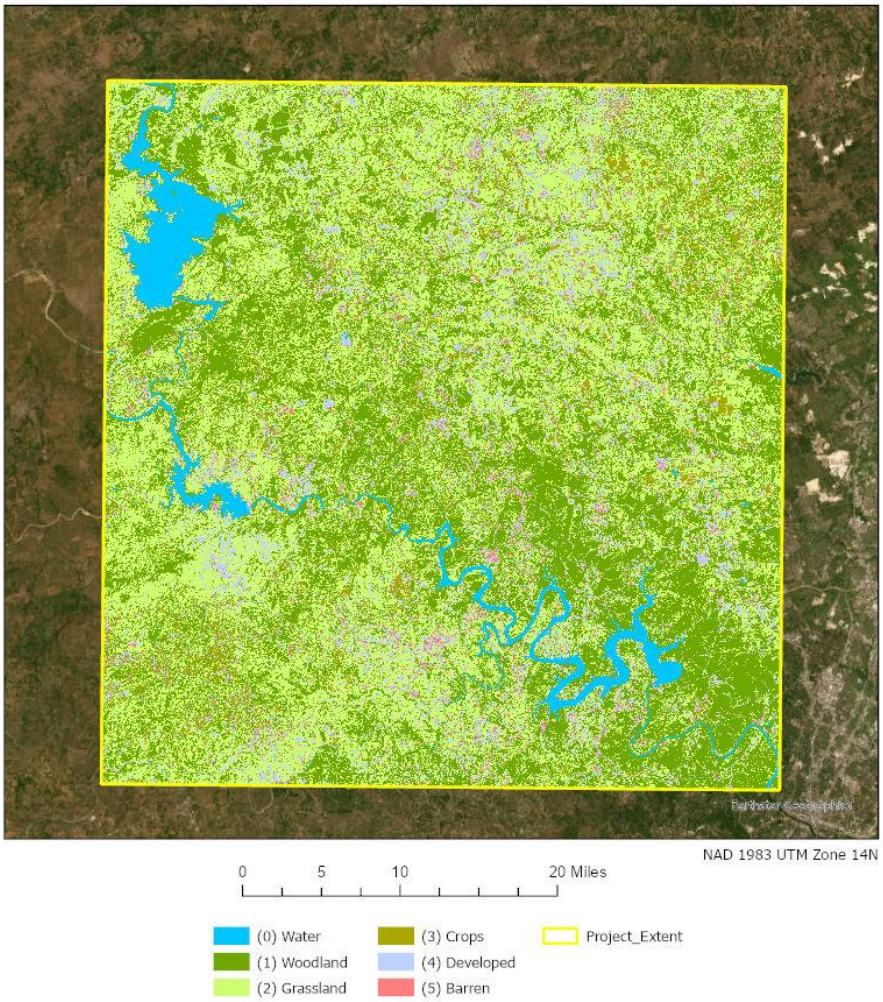


Figure 4

Highland Lakes 1987

Unsupervised Classification

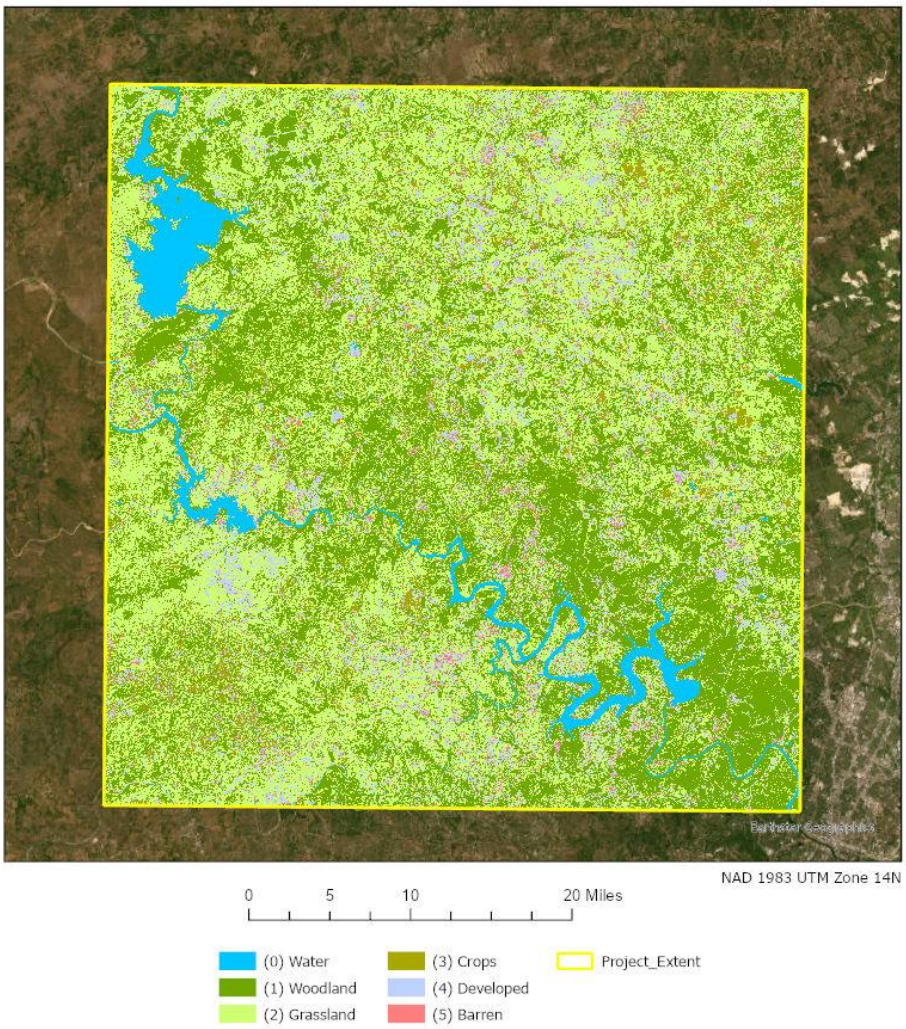


Figure 5



Accuracy Assessment

The accuracy assessment for both the supervised and unsupervised models are disappointing and may warrant another analysis to see if there are better results. As stated earlier, the overall accuracy of the supervised classification for 1987 is 53.3% and for 2011, it's 40.0%. For the overall accuracy of the supervised classification for 1987 is 33.3% and for 2011, it's 43.3%.

Model	OBJECTID	Class	ClassValue	C_0	C_1	C_2	C_3	C_4	C_5	Total	U_Accuracy	Kappa
ConfusionMatrix_LC_1987_SC	1	Water	C_0	3	0	0	0	0	0	3	1	0
Supervised Classificaton 1987	2	Woodland	C_1	0	2	3	0	2	0	7	0.285714286	0
	3	Grassland	C_2	0	0	1	0	0	0	1	1	0
	4	Crops	C_3	0	1	5	1	0	1	8	0.125	0
	5	Developed	C_4	0	3	0	0	3	1	7	0.428571429	0
	6	Barren	C_5	1	1	2	0	0	0	4	0	0
	7		Total	4	7	11	1	5	2	30	0	0
	8		P_Accuracy	0.75	0.285714286	0.090909091	1	0.6	0	0	0.333333333	0
	9		Kappa	0	0	0	0	0	0	0	0	0.227799228

Model	OBJECTID	Class	ClassValue	C_0	C_1	C_2	C_3	C_4	C_5	Total	U_Accuracy	Kappa
ConfusionMatrix_LC_2011_SC	1	Water	C_0	2	0	0	0	0	0	2	1	0
Supervised Classification 2011	2	Woodland	C_1	0	4	0	0	1	0	5	0.8	0
	3	Grassland	C_2	0	0	2	0	0	0	2	1	0
	4	Crops	C_3	0	0	3	1	0	1	5	0.2	0
	5	Developed	C_4	2	3	6	0	4	1	16	0.25	0
	6	Barren	C_5	0	0	0	0	0	0	0	0	0
	7		Total	4	7	11	1	5	2	30	0	0
	8		P_Accuracy	0.5	0.571428571	0.181818182	1	0.8	0	0	0.433333333	0
	9		Kappa	0	0	0	0	0	0	0	0	0.32

Model	OBJECTID	Class	ClassValue	C_0	C_1	C_2	C_3	C_4	C_5	Total	U_Accuracy	Kappa
ConfusionMatrix_LC_1987_UC	1	Water	C_0	3	0	0	0	0	0	3	1	0
Unsupervised Classification 1987	2	Woodland	C_1	0	4	1	0	0	0	5	0.8	0
	3	Grassland	C_2	1	2	5	0	2	1	11	0.454545455	0
	4	Crops	C_3	0	0	4	0	0	0	4	0	0
	5	Developed	C_4	0	1	1	1	3	0	6	0.5	0
	6	Barren	C_5	0	0	0	0	0	1	1	1	0
	7		Total	4	7	11	1	5	2	30	0	0
	8		P_Accuracy	0.75	0.571428571	0.454545455	0	0.6	0.5	0	0.533333333	0
	9		Kappa	0	0	0	0	0	0	0	0	0.396551724

Model	OBJECTID	Class	ClassValue	C_0	C_1	C_2	C_3	C_4	C_5	Total	U_Accuracy	Kappa
ConfusionMatrix_LC_2011_UC	1	Water	C_0	2	0	0	0	0	0	2	1	0
Unsupervised Calssification 2011	2	Woodland	C_1	0	5	3	0	1	0	9	0.555555556	0
	3	Grassland	C_2	0	0	2	1	1	1	5	0.4	0
	4	Crops	C_3	0	1	3	0	0	1	5	0	0
	5	Developed	C_4	2	1	3	0	3	0	9	0.333333333	0
	6	Barren	C_5	0	0	0	0	0	0	0	0	0
	7		Total	4	7	11	1	5	2	30	0	0
	8		P_Accuracy	0.5	0.714285714	0.181818182	0	0.6	0	0	0.4	0
	9		Kappa	0	0	0	0	0	0	0	0	0.254143646

Conclusion

Based on the accuracy assessment, the overall accuracy was very similar between supervised and unsupervised classification with a slight advantage going towards unsupervised classification. The process of classifying all 30 training samples was challenging by going back and forth between the Landsat imagery. With a resolution of 30 meters, this is very difficult to do, so the ArcGIS World Imagery was used with it's one meter resolution. The problem is that this is current and what we are analyzing is either 1987 or 2011. Later, it is shown through error matrix that this will affect the producer's and user's accuracy since these sample data points will be used as the training or reference data. A couple of ways to address this would be to follow the Statement of Work (SOW)'s suggestion of using Landsat 5 images and experimenting with a variety of band combinations. Also, the SOW suggested using USGS NLCD and TPWD Ecological Mapping Systems land cover datasets to help as well. In all, despite the issues going through the

process, the main point of the study to see the change between 1987 and 2011 was clear. More iterations will improve the process with time.